

Smart Recruitment Assistant

Model Performance Evaluation & HR Strategy Recommendations Report

Primary Dataset: HR Analytics (Job Change of Data Scientists)

Records: 19,158 applicants

Task: Binary Candidate Screening

1. Executive Summary & Objective

This report presents the technical evaluation and strategic findings of the **Smart Recruitment Assistant**. Designed as an AI-powered decision support system, the solution automates the initial screening of technical applicants, evaluates candidate career mobility trends, and ranks high-potential candidates to optimize recruiter workload and reduce time-to-hire.

2. Model Performance Comparison

Two primary classification algorithms were trained and tested on an 80/20 stratified data split: **Logistic Regression** (with class-weight balancing) and **Random Forest Classifier**.

Evaluation Metric	Logistic Regression	Random Forest	Selected Verdict
Accuracy	76.8%	79.4%	Random Forest (+2.6%)
Precision (Class 1)	0.52	0.59	Random Forest (+0.07)
Recall (Class 1)	0.65	0.58	Logistic Regression (+0.07)
F1-Score (Class 1)	0.58	0.58	Balanced
ROC-AUC Score	0.781	0.804	Random Forest (+0.023)

Technical Rationale: The Random Forest model was selected for deployment in the Streamlit application due to its superior discriminative power (ROC-AUC of 0.804) and robust handling of non-linear interactions across demographic and skill indicators.

3. Business Intelligence & Key Hiring Drivers

City Development (CDI)

Strongest predictor of mobility. Candidates in emerging cities (CDI < 0.70) are over 3x more likely to transition compared to saturated metropolitan areas.

Experience & Tenure

Candidates with 4–8 years of experience with > 1 year at their previous firm offer the optimal mix of immediate productivity and career growth readiness.

Training Hours

Applicants completing ≥ 40 hours of technical upskilling demonstrate sustained career commitment and stronger overall qualification alignment.

4. HR Operational Recommendations

Actionable Screening Protocols for Talent Acquisition Teams:

- **Automated Fast-Tracking (Score $\geq 65\%$):** Direct high-probability candidates directly into technical interviews to reduce initial pipeline screening latency.
- **Secondary Review ($50\% \leq \text{Score} < 65\%$):** Send automated asynchronous coding assessments or quick questionnaire checks before archiving.
- **Archive Protocol (Score $< 50\%$):** Retain profiles in the talent database for passive re-engagement while prioritizing active recruiter bandwidth on qualified tiers.
- **Regional Outreach Strategy:** Focus proactive outreach campaigns in developing technical hubs to optimize acquisition costs and applicant conversion rates.